

# Making NDFU bottom-up: the unified adoption rule and its two variants

NDFU internal design note

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Status: design document — no changes to the engine have been made.

Companion scripts: `10_unified_model_sweep.R`, `12_unified_premium_sweep.R`

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## 1 Why this document

The Social Networks special-issue rejection (Bianchi’s follow-up email) said:

*“the underdevelopment of the behavioural model assumed in your ABM. While it can certainly be considered an agent-based model, the behaviour of the agents is relatively*

*abstract, and the experimental design appears to be governed by macro-level manipulation rather than bottom-up dynamics.”*

Two pieces of the current model are exposed to that critique:

1. **The network threshold  $\tau_i$  is exogenous.** It is drawn from  $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ , truncated to  $[0, 1]$ , *independently of the agent’s own measured preference  $q_i$* . The sweep over  $(\tau_{\text{mean}}, \tau_{\text{sd}})$  is exactly the “macro-level manipulation” Bianchi points at: a population-level dial with no micro-foundation in the survey data.
2. **The two channels are bolted together.** Rational adoption ( $q_i \leq \Gamma$ ) and social adoption ( $\tau_i \leq \tilde{E}_i$ ) are two separate rules joined by an OR. A referee can ask why an agent runs two unrelated decision procedures.

Tur, Zeppini & Frenken (2024, *Social Networks* 76:12–21) — published in our target journal — has a *single* micro rule (a utility comparison, where social reinforcement enters the utility calculus), but **no social-distance sensitivity in influence**: every adopting neighbor reinforces equally. NDFU’s distinctive ingredient is exactly that sensitivity (Centola 2011). The question is therefore:

Can we keep NDFU’s selective (distance-sensitized) influence while adopting a Tur-style single utility-based decision rule — making the model bottom-up without giving up its identity?

Answer: yes. The instrument is a single **unified adoption rule** (§4), which admits two operationalizations — a *linear-fractional* variant and a *multiplicative  $\gamma$ -variant* that nests Tur (2024) exactly. Their pros and cons are compared in detail in §5.

## 2 Notation and the current NDFU model (baseline)

| Symbol              | Meaning                                                     | Source                                                               |
|---------------------|-------------------------------------------------------------|----------------------------------------------------------------------|
| $\Gamma \in [0, 1]$ | Intrinsic Utility Level (IUL) of the behavior               | swept                                                                |
| $q_i \in [0, 1]$    | Minimum Utility Requirement (MUR): intrinsic aversion       | <b>empirical</b> (GSS/ATP items)                                     |
| $\tau_i \in [0, 1]$ | network threshold (fraction of influential adopters needed) | <b>exogenous</b> $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ |
| $d_{ij} \in [0, 1]$ | Gower distance in Blau space (age, educ, race, relig, sex)  | <b>empirical</b>                                                     |
| $h \in [0, 1]$      | Maximum Social Proximity (MSP): tolerance to dissimilarity  | swept (macro)                                                        |
| $a_j \in \{0, 1\}$  | adoption state of $j$                                       | dynamic                                                              |
| $X_{ij}$            | adjacency                                                   | imputed ERGM networks                                                |

### Current rules (as in the paper).

- Rational channel: adopt if  $q_i \leq \Gamma$  (Eq. 1 of the paper).
- Social channel: adopt if  $\tau_i \leq \tilde{E}_i$ , with selective effective exposure

$$\tilde{E}_i = \frac{\sum_{j \neq i} X_{ij} \tilde{a}_j}{\sum_{j \neq i} X_{ij}}, \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(h - d_{ij}), \quad (1)$$

where  $U_{ij} \sim \text{Unif}(0, 1)$  is redrawn per step and  $\sigma(z) = (1 + e^{-z/T})^{-1}$  with  $T = 0.02$  in the implementation (a near-hard gate at  $d_{ij} = h$ ; see `scripts/04_GSS_diffusion_sims.R`, line 304).

**Implementation nuance worth knowing** (it matters for the comparison with Tur): in the code, the rational channel is implemented by setting the node’s threshold to  $10^{-6}$ , so a rational agent still needs **one influential contact** (exposure  $> 0$ ) before adopting. The engine is therefore already a word-of-mouth model for *awareness*: low- $q$  agents behave as simple contagion, not as spontaneous ( $\varphi = 0$ ) adopters. The paper’s Eq. 1 (adoption at exposure 0) is the idealization; the code is contact-gated.

### 3 Tur (2024) mechanism, for reference

- Word-of-mouth **percolation**: agents must be *informed* through the network before they can decide.
- Single decision rule: adopt when the innovation’s intrinsic utility  $\xi$  meets the agent’s minimum utility requirement.
- **Social reinforcement modifies the requirement** — exact published form (confirmed against both PDFs):

$$u_t^i = u_0^i \cdot \left( \frac{1}{a_t^i} \right)^\gamma, \quad (2)$$

where  $u_0^i$  is the initial minimum utility requirement,  $a_t^i \in \mathbb{N}$  the number of **adopting neighbors** of  $i$  at time  $t$  (a state count, not received recommendations; the rule is evaluated only once informed, so  $a \geq 1$ ), and  $\gamma \geq 0$  the social-reinforcement strength. The lowering is **multiplicative** (power-law): each doubling of adopting neighbors divides the requirement by the same factor  $2^\gamma$ . With  $\gamma = 0$  the model reduces to pure social percolation. The threshold-like behavior emerges from this lowering, not from a separate threshold parameter.

- Homophily  $\rho$  is purely **structural pre-wiring**: agents with similar  $u$  are placed together *before* diffusion. Influence itself is **distance-blind**: any adopting neighbor reinforces equally.
- Preferences are artificial distributions (Beta( $\alpha, \beta$ ) family in 2018; uniform in 2024); networks are Watts–Strogatz with  $N = 10,000$  and  $k = 4$  neighbors on average.

NDFU vs. Tur, in one line: **Tur has the single bottom-up rule but uniform influence; NDFU has selective influence but a two-channel rule with an exogenous  $\tau$** . Each has what the other lacks.

## 4 The unified additive-utility rule

### 4.1 The rule

Replace both channels by one utility comparison per agent per step:

$$\text{adopt}_i(t) = 1 \iff \Gamma + \lambda \tilde{E}_i(t) \geq q_i. \quad (3)$$

- $\tilde{E}_i(t)$  is **exactly the current selective effective exposure** (1), unchanged. This is where social distance enters: only adopters passing the stochastic proximity gate  $\sigma(h - d_{ij})$  count.
- $\lambda > 0$  is the **social-proof coupling**: how much situational utility a fully-adopted, fully-similar neighborhood adds, in the same units as  $\Gamma$  and  $q_i$ .

### 4.2 Two algebraically identical framings (choose the second)

1. **Requirement-lowering (Tur’s framing)**:  $q_i^{\text{eff}}(t) = q_i - \lambda \tilde{E}_i(t)$ ; adopt when  $\Gamma \geq q_i^{\text{eff}}(t)$ . Preferences are *modified* during diffusion.

2. **Situational-utility (recommended framing):**  $U_i(t) = \Gamma + \lambda \tilde{E}_i(t)$ ; adopt when  $U_i(t) \geq q_i$ . The preference  $q_i$  **never changes**; what changes is the utility the situation offers (peers add legitimacy, lower uncertainty, network externalities). This is literally Granovetter’s threshold as a *situated cost–benefit crossing* (1978, p. 1422), and it preserves NDFU’s declared contrast with Tur (“preferences are not modified during diffusion”, `theoretical_reviewed.md`, entry 7).

### 4.3 The derived threshold — the two old channels fall out as limit cases

Rearranging (3), adoption requires  $\tilde{E}_i(t) \geq \tau_i^*$  with the **derived threshold**

$$\tau_i^* = \max\left\{0, \frac{q_i - \Gamma}{\lambda}\right\}. \quad (4)$$

- $q_i \leq \Gamma \Rightarrow \tau_i^* = 0$ : adopts at first influential contact — **the current rational channel** (Granovetter’s low-threshold “instigator”, exactly as currently implemented with  $\tau = 10^{-6}$ ).
- $q_i > \Gamma \Rightarrow \tau_i^* > 0$ : needs selective exposure proportional to the **utility deficit** — the social channel, now derived instead of postulated.
- $q_i > \Gamma + \lambda \Rightarrow \tau_i^* > 1$ : **can never adopt** ( $\tilde{E} \leq 1$ ) — a “structurally immune” agent. When exactly this occurs is worked out in §5.1.

### 4.4 What this buys and what it costs

**Buy.**

- **Bottom-up microfoundation.** One decision rule, all heterogeneity empirical:  $\tau_i^*$  is a deterministic function of the *measured*  $q_i$  and the product  $(\Gamma, \lambda)$ . The arbitrary  $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$  disappears. This is the direct answer to Bianchi.
- **Behavioral realism as a bonus:** resistance to peer influence is now positively correlated with intrinsic aversion (the same latent trait drives both) — defensible, and testable in principle.
- **Nesting.** Setting the gate to  $w \equiv 1$  ( $h \rightarrow \infty$ , distance-blind) removes the distance selectivity and leaves a Tur-style (distance-blind) reinforcement model; setting  $\lambda \rightarrow 0$  recovers a pure percolation-of-awareness model. Tur (2024)’s *exact* published form (2) is recovered by the  $\gamma$ -variant of the rule — see §4.6. NDFU’s selectivity remains the *identified added ingredient*, cleanly positioned against a paper published in the target journal.
- **Implementability.**  $\tau_i^*$  is static given  $(\Gamma, \lambda)$ , so the existing `netdiffuseR` threshold machinery runs it as-is: `threshold.dist = pmax(0, (q - Γ)/λ)` replaces the Normal draw + the  $10^{-6}$  hack. No engine changes needed.

**Costs / changes.**

- The rational/social decomposition becomes **ex-post bookkeeping** (adopters with  $q_i \leq \Gamma$  vs.  $q_i > \Gamma$ ) rather than two mechanisms — same tally we already compute, but the interpretation shifts.
- $\tau_{\text{sd}}$  disappears as a free dial; its large positive effect on cascades (heterogeneity creates susceptible minorities) is now carried by the *empirical* dispersion of  $q_i$ , scaled by  $1/\lambda$ .  $\lambda$  replaces  $(\tau_{\text{mean}}, \tau_{\text{sd}})$  — the parameter space shrinks from 16 ( $4 \times 4$ ) threshold configurations to a 1-D  $\lambda$  sweep.
- The (IUL  $\times$  MSP) heatmaps will not be identical: thresholds now co-vary with  $q$ . Whether the phase structure survives is an empirical question — answered: it does (see §7).

## 4.5 Where social distance enters (explicitly)

Only inside  $\tilde{E}_i$ , via the per-contact stochastic gate:

$$\Pr(j\text{'s adoption counts for } i) = \sigma\left(\frac{h - d_{ij}}{T}\right), \quad T = 0.02. \quad (5)$$

Identical to the current model. The unified rule changes the *decision rule*, not the *influence filter*.

## 4.6 Recovering Tur (2024) exactly — the nesting, step by step

**Tur’s published rule** (exact form, §3): an agent that has been *informed* through the network adopts when the product’s utility meets its *current* requirement,

$$\xi \geq u_t^i, \quad u_t^i = u_0^i \cdot \left(\frac{1}{a_t^i}\right)^\gamma, \quad (6)$$

with  $u_0^i$  the initial minimum utility requirement,  $a_t^i \geq 1$  the number of adopting neighbors of  $i$  at time  $t$  (the rule is only evaluated once informed, so  $a \geq 1$  and the expression is well-defined), and  $\gamma \geq 0$  the social-reinforcement strength.  $\gamma = 0$  gives  $u_t = u_0$ : pure social percolation (Solomon & Weisbuch 2000), Tur’s own baseline.

**Log-linearization — Tur IS an additive requirement-lowering rule, in logs.** Taking logarithms (all quantities positive):

$$\ln \xi \geq \ln u_0^i - \gamma \ln a_t^i. \quad (7)$$

So the published reinforcement is additive with the concave gain  $g(a) = \ln a$ . This dictates the right unified rule: it must be additive in *log-utility*, i.e. **multiplicative in levels**.

**The Tur-exact unified rule ( $\gamma$ -form).** Define the distance-sensitized adopting *count*

$$\tilde{A}_i(t) = \sum_j X_{ij} \tilde{a}_j(t), \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma\left(\frac{h - d_{ij}}{T}\right), \quad (8)$$

and let social proof *multiply* the situational utility:

$$\boxed{\text{adopt}_i(t) \iff \Gamma \cdot (\tilde{A}_i(t))^\gamma \geq q_i} \quad (\text{evaluated when } \tilde{A}_i \geq 1). \quad (9)$$

Equivalently, in Tur’s own requirement-lowering form:  $q_{i,t} = q_i \cdot (1/\tilde{A}_i(t))^\gamma$ , adopt when  $\Gamma \geq q_{i,t}$ . The preference  $q_i$  is fixed; what the similar adopters change is the *situational* requirement — the same framing as §4.2 (but see §5.5 for the substantive difference this form does carry).

**Recovery, in one step.** Switch off the proximity gate ( $h \geq 1 + 5T \Rightarrow \sigma \rightarrow 1 \Rightarrow \tilde{a}_j = a_j$ , so  $\tilde{A}_i(t) = a_t^i$ ), and rule (9) reads

$$\Gamma \cdot (a_t^i)^\gamma \geq q_i \iff \xi \geq u_0^i \cdot \left(\frac{1}{a_t^i}\right)^\gamma \quad (10)$$

under the identification

$$\boxed{\xi = \Gamma, \quad u_0^i = q_i, \quad \text{same } \gamma.} \quad (11)$$

The recovery is **exact on any graph**: Tur’s reinforcement is a pure count, and the  $\gamma$ -form uses the count directly — no degree normalization is involved, so no degree-regularity caveat is needed (unlike the linear-fractional variant, which normalizes by degree and matches Tur’s count reading only on degree-regular substrates).

**Awareness / percolation layer** (unchanged): the engine’s contact-gating (the  $\tau = 10^{-6}$  device, §2) means the rule is only evaluated at  $\tilde{A}_i \geq 1$  — precisely Tur’s “informed” condition with per-contact transmission probability 1; a probability  $p < 1$  is emulated by a constant-acceptance gate ( $\sigma \equiv p$ ).

**Derived threshold (count form).** Rearranging (9):

$$\text{adopt}_i \iff \tilde{A}_i(t) \geq c_i^*, \quad c_i^* = \left(\frac{q_i}{\Gamma}\right)^{1/\gamma}. \quad (12)$$

- $q_i \leq \Gamma \Rightarrow c_i^* \leq 1$ : adopts at the first influential contact — the rational channel, exactly as in the linear variant (§4.3).
- $q_i > \Gamma \Rightarrow$  needs  $c_i^*$  similar adopters; heterogeneity 100% empirical (from  $q_i$ ), with  $\gamma$  as the single coupling.
- **Degree-dependent immunity**: agent  $i$  is out of reach whenever  $c_i^* > k_i$  (its degree) — high-aversion, *low-degree* agents can never be socially converted (see §5.1).

**What does NOT nest (honesty box).**

- Tur’s homophily  $\rho$  sorts agents by the requirement *u itself*; NDFU’s imputation sorts by demographics, and  $q_i$  correlates with demographics only partially. The imputed networks are therefore an *empirically disciplined, weaker* version of Tur’s preference-sorting. The nesting claim concerns the **adoption rule**; the network is an exogenous input in both models.

**One-line summary.** With the proximity gate switched off, the  $\gamma$ -form is Tur (2024) — same functional form, same  $\gamma$ , with  $\xi = \Gamma$  and  $u_0 = q$  — on any topology; NDFU’s added ingredient is exactly one: the distance gate deciding *which* adopters count in  $\tilde{A}_i$ .

## 5 Pros and cons of the two operationalizations

We now have two members of one family on the table:

$$\textbf{Linear-fractional:} \quad \Gamma + \lambda \tilde{E}_i(t) \geq q_i \quad (\text{additive in the adopted } \textit{fraction}) \quad (13)$$

$$\textbf{\gamma-form (Tur-exact):} \quad \Gamma \cdot \tilde{A}_i(t)^\gamma \geq q_i \quad (\text{multiplicative in the adopted } \textit{count}) \quad (14)$$

This section works through the five questions that decide between them, pedagogically and one at a time, and closes with a verdict table.

### 5.1 When can an agent never adopt? (immunity, made precise)

In the linear variant, immunity is  $\tau_i^* > 1$ , i.e.

$$q_i > \Gamma + \lambda. \quad (15)$$

Since  $q_i \leq 1$ , this region is **non-empty only when**  $\lambda < 1 - \Gamma$ : with  $\lambda \geq 1$  immunity is impossible (a fully-adopted similar neighborhood can compensate any deficit). So the intuition “ $q_i, \Gamma \in [0, 1]$  makes this very unlikely” is correct *at large*  $\lambda$  but not in general. Empirically, on the GSS  $q$  distribution (diagnostic printed by `10_unified_model_sweep.R`):

| $\lambda$      | $\Gamma$ | immunity condition              | share immune |
|----------------|----------|---------------------------------|--------------|
| 0.5            | 0.20     | $q_i > 0.70$                    | 10.9%        |
| 0.5            | 0.30     | $q_i > 0.80$                    | 3.8%         |
| 1.0–2.0        | any      | $q_i > \Gamma + \lambda \geq 1$ | 0%           |
| 0.7 (best fit) | 0.20     | $q_i > 0.90$                    | small tail   |

The visible fingerprint: the adoption **ceiling**  $\Phi \approx 0.75$  at low IUL in the  $\lambda = 0.5$  heatmaps — the  $\approx 11\%$  immune tail (plus isolates) caps the ordered phase below 1. This is a prediction, not a bug: *when social proof is weak, its reach is bounded*. The current engine cannot produce this ceiling because  $\tau$  is truncated to  $[0, 1]$  by construction.

In the  $\gamma$ -form, immunity is instead  $c_i^* > k_i$ , i.e.

$$\left(\frac{q_i}{\Gamma}\right)^{1/\gamma} > k_i \iff q_i > \Gamma k_i^\gamma, \quad (16)$$

which depends on the agent’s **degree**: the out-of-reach agents are those both highly averse *and* poorly connected. Arguably more realistic — and a distinctive, testable difference between the two variants.

## 5.2 The log-space bridge: $\ln a$ vs. $\tilde{E}$

In log space, Tur’s reinforcement variable is  $\ln a$  — the log of the gated *count*. Is that “just” our normalized exposure? Almost: with  $\tilde{E}_i = \tilde{A}_i/k_i$ ,

$$\ln \tilde{A}_i = \ln \tilde{E}_i + \ln k_i. \quad (17)$$

On **degree-regular substrates** — and Tur’s Watts–Strogatz networks are degree-homogeneous with  $k = 4$  — the term  $\ln k_i$  is one global constant, absorbable into  $\Gamma$  (or  $u_0$ ): identifying the reinforcement with  $\ln \tilde{E}$  is then exact up to recalibration. On NDFU’s **heterogeneous** networks (degrees roughly 1–80), the two differ by the agent-specific offset  $\gamma \ln k_i$ :

- using  $\tilde{E}$  says “reinforcement is a *share of your attention*”;
- using  $\tilde{A}$  says “every endorsement adds the same, whether you have 4 neighbors or 40” — at the same adopted *fraction*, a hub receives more reinforcement.

This is the count-vs-fraction fork of §4.6, seen from the logarithm: the analogy  $\ln a \leftrightarrow \tilde{E}$  is complete *modulo degree*, and the choice between the two variants is precisely the choice of what to do with  $\ln k_i$ .

## 5.3 What $\gamma$ is — and is not — in Tur 2018 and Tur 2024

Read against both PDFs, the epistemic status of  $\gamma$  is the following.

**What their justification does have.**

- An **axiomatic characterization** (2018, p. 3; restated in 2024): the updated requirement  $f(q; a, \gamma) = q \cdot (1/a)^\gamma$  must satisfy (a) decreasing in  $a$  for  $\gamma > 0$ ; (b) decreasing in  $\gamma$  for  $a > 1$ ; (c)  $\gamma = 0$  replicates the percolation benchmark; (d)  $a = 1$  gives  $f = q$  (the *first* adopting neighbor only informs; reinforcement starts with the second — a clean echo of complex contagion’s “more than one source”).
- A **concavity argument** for the restriction  $\gamma \leq 1$ : “assuming that the marginal effect of each additional neighbor on the total social influence cannot be increasing” (2018), with the reductio that a very large  $\gamma$  trivializes diffusion *on their degree-4 lattices*.
- An interpretable **marginal effect** (2018, Eq. 2):  $\Delta q_t^i = -q_0^i \gamma / (a_t^i)^{\gamma+1} \cdot \Delta a$  — the first adopting friends produce the large drops, and the drop is *proportional to the agent’s own requirement* (more averse agents receive larger absolute reductions). This proportionality is the seed of the complementarity discussed in §5.5.



## What it does not have.

- **No empirical calibration.** In both papers  $\gamma$  is a free “intensity” parameter, introduced with one sentence (2024: “for some innovations, social reinforcement may be more intense than for others . . . so we introduce social reinforcement as a parameter”) and *swept*:  $\gamma \in [0, 1]$  continuously in the 2018 phase maps and the 2024 heat maps, with focal values  $\{0, 0.5, 1\}$  in 2024’s Fig. 5 (“ $\gamma = 1$  is maximum social reinforcement”). There are no units, no measurement strategy, no experimental anchor — not even a link to Centola’s dose–response evidence, which would have been the natural one.
- **No transportability across degree scales.** All Tur simulations live on  $k = 4$  neighborhoods, so  $a \in \{1, \dots, 4\}$  in practice and the reinforcement spans at most a factor  $4^\gamma \leq 4$ . On NDFU’s imputed networks (mean degree  $\approx 29$ , hubs beyond 60), the same  $\gamma$  produces reductions up to  $29^\gamma$ : at  $\gamma = 1$  a saturated average neighborhood divides the requirement by  $\approx 29$ , a force of a different order of magnitude. Even their  $\gamma \leq 1$  bound is reasoned *for*  $k = 4$ . **Borrowing Tur’s  $\gamma$  values would inherit the symbol, not the calibration.**

**Verdict.** Adopting the  $\gamma$ -form buys (i) exact formal nesting, (ii) the published axioms and the concavity argument for free, and (iii) a citation in the target journal. It does *not* buy empirical values for  $\gamma$ , nor freedom from arguing the coupling’s magnitude — if the  $\gamma$ -form is used,  $\gamma$  must be calibrated for *our* degree distribution (e.g. with the same fit-to-current-model procedure used for  $\lambda$ ). Epistemically,  $\gamma$  and  $\lambda$  are in the same place: free couplings — with the difference that  $\lambda$  has the stronger microeconomic backing (§5.6).

## 5.4 Exposure traditions: the power form is wedded to counts

The  $\gamma$ -form cannot be written with normalized exposure, and the reason is structural, not stylistic: the multiplier  $\tilde{A}^\gamma$  amplifies only because  $\tilde{A} \geq 1$ . Substituting the fraction  $\tilde{E} \in [0, 1]$  gives  $\tilde{E}^\gamma \leq 1$  — social proof would *reduce* situational utility. There is no natural “fractional  $\gamma$ -form”.

Consequences:

- Choosing the  $\gamma$ -form moves the model from the fractional-threshold tradition (Granovetter 1978; Valente 1996; Watts 2002) — the tradition NDFU and `netdiffuseR` live in — to the absolute-count tradition (Centola & Macy 2007). For a *Social Networks* audience this is not neutral.
- **Mitigation 1 (mechanism  $\neq$  measurement):** even if the decision rule runs on counts, Valente-style exposure-at-adoption can still be computed and reported *ex post* as a descriptive statistic (measured thresholds). Nothing forbids reporting  $\tilde{E}_i$  at the moment of adoption.
- **Mitigation 2 (tooling):** `netdiffuseR` supports non-normalized exposure (`exposure()` with `normalized = FALSE`), so the count threshold  $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$  runs on the same machinery. The loss is conceptual continuity, not implementation.

## 5.5 Are preferences modified? Composition, not units

Formally, the  $\gamma$ -form does *not* force the “preferences change” reading:  $\Gamma \cdot \tilde{A}^\gamma \geq q_i$  is symmetric, and the factor  $\tilde{A}^\gamma$  can be booked on either side —

$$\underbrace{U_i(t) = \Gamma \cdot \tilde{A}_i^\gamma \geq q_i}_{\text{situational utility multiplied; preference fixed}} \iff \underbrace{\Gamma \geq q_i \cdot \tilde{A}_i^{-\gamma} = q_{i,t}}_{\text{requirement lowered (Tur’s framing)}} \quad (18)$$

with identical behavior. In logs the first reading is a Cobb–Douglas-style utility in two “goods” (product quality, social proof):  $\ln U_i = \ln \Gamma + \gamma \ln \tilde{A}_i$ . Both sides live on the same  $[0, 1]$  utility scale as  $q_i$ , so the real distinction is not one of *units* but of **composition**: additive vs. multiplicative.



And the composition carries substance. Compare the marginal value of social proof:

$$\text{linear:} \quad U_i = \Gamma + \lambda \tilde{E}_i, \quad \frac{\partial U_i}{\partial \tilde{E}_i} = \lambda, \quad \frac{\partial^2 U_i}{\partial \Gamma \partial \tilde{E}_i} = 0 \quad (\text{substitutes}), \quad (19)$$

$$\gamma\text{-form:} \quad U_i = \Gamma \cdot \tilde{A}_i^\gamma, \quad \frac{\partial U_i}{\partial \tilde{A}_i} = \gamma \Gamma \tilde{A}_i^{\gamma-1}, \quad \frac{\partial^2 U_i}{\partial \Gamma \partial \tilde{A}_i} = \gamma \tilde{A}_i^{\gamma-1} > 0 \quad (\text{complements}). \quad (20)$$

In the linear form, a given (similar) adopting neighborhood contributes the same  $\lambda \tilde{E}$  regardless of how good the product is: quality and social proof are *substitutes* (additively separable), and the social contribution needed to convert agent  $i$  compensates the utility deficit one-for-one,  $\lambda \tilde{E}_i \geq q_i - \Gamma$ , independently of  $\Gamma$ 's level. In the  $\gamma$ -form the cross-partial is strictly positive (complementarity in the Edgeworth–Pareto sense): *the same adopting neighborhood is worth more for better products*. The mirror statement on the requirement side is Tur 2018's own Eq. 2, where the drop is proportional to  $q_0^i$ . The required social proof falls steeply with quality,

$$\ln c_i^* = \frac{\ln q_i - \ln \Gamma}{\gamma}, \quad \frac{d \ln c_i^*}{d \ln \Gamma} = -\frac{1}{\gamma}, \quad (21)$$

an elasticity of  $-1/\gamma$  — e.g. at  $\gamma = 0.5$ , doubling the product's quality divides the required count of similar adopters by four.

So the honest summary of the user's point: it *can* be framed as preferences-fixed either way, but the multiplicative interaction is a substantive behavioral commitment (quality–social-proof complementarity), not just bookkeeping — and since Tur themselves present the form as requirement-lowering, adopting their exact form and citation invites the “preferences change” reading unless a paragraph defends the other bookkeeping.

## 5.6 The Brock–Durlauf microfoundation of the linear form

The linear variant is not ad hoc: it is the deterministic limit of the canonical discrete-choice-with-social-interactions model of Brock & Durlauf (2001, *Review of Economic Studies* 68(2):235–260). In their setup, agent  $i$  chooses a binary action and derives utility

$$V_i(\omega_i) = \underbrace{h_i(\omega_i)}_{\text{private utility}} + \underbrace{J \omega_i \bar{m}_i^e}_{\text{social utility (conformity)}} + \underbrace{\varepsilon_i(\omega_i)}_{\text{random shock}}, \quad (22)$$

where  $\bar{m}_i^e$  is the expected mean choice of  $i$ 's *reference group* and  $J \geq 0$  is the social-interaction (conformity) coefficient. Three points of contact:

1. **Additive separability of private and social utility is the standard assumption** of that literature — exactly the structure of the linear rule. Mapping: the private net utility of adopting is  $\Gamma - q_i$ ; the social term is  $\lambda \tilde{E}_i$ , with  $\tilde{E}_i$  playing  $\bar{m}_i^e$  (the adoption rate of the reference group) and  $\lambda$  playing  $J$ . Adoption when total utility is non-negative:  $\Gamma - q_i + \lambda \tilde{E}_i \geq 0$  is (3) rearranged.
2. **NDFU's additions are well-defined deviations, not violations:** the reference group is not exogenous but *similarity-gated* (the  $\sigma(h - d_{ij})$  filter selects who belongs to  $i$ 's effective reference group — our contribution), and the shock is degenerate ( $\varepsilon \rightarrow 0$ ), turning Brock–Durlauf's logit choice probabilities into a deterministic threshold rule. Reintroducing a small  $\varepsilon$  would give a smooth (logit) adoption rule — a natural robustness variant.
3.  **$J$  has an econometric life that  $\gamma$  lacks.** The Brock–Durlauf coefficient underpins the social-multiplier / social-interactions estimation literature (see Blume, Brock, Durlauf & Ioannides 2011, *Handbook of Social Economics*, for identification), so  $\lambda$  inherits a path toward empirical anchoring. As a bonus, Brock & Durlauf themselves develop the mean-field correspondence of (22) with the Ising model — which connects this microfoundation directly to NDFU's criticality analysis.

## 5.7 Verdict table and recommendation

|                               | <b>Linear-fractional</b> — $\Gamma + \lambda \tilde{E}_i \geq q_i$                                                        | <b><math>\gamma</math>-form (count)</b> — $\Gamma \cdot \tilde{A}_i^\gamma \geq q_i$                                                      |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Answer to Bianchi             | ✓ equally bottom-up (threshold derived from measured $q$ )                                                                | ✓ equally bottom-up                                                                                                                       |
| Backing of the coupling       | $\lambda$ with Brock–Durlauf microfoundation (additive private + social utility; §5.6); its magnitude still needs arguing | $\gamma$ with Tur precedent (axioms + concavity), but <b>no empirical calibration and no transportability</b> across degree scales (§5.3) |
| Nesting in Tur 2024           | only qualitative                                                                                                          | <b>exact, on any graph</b> (gate off; $\xi=\Gamma$ , $u_0=q$ , same $\gamma$ )                                                            |
| Exposure tradition            | ✓ Valente/Granovetter fractional; <b>netdiffuseR</b> native                                                               | counts (Centola–Macy/Watts); Valente exposure only as ex-post reporting (§5.4)                                                            |
| Preferences-fixed framing     | ✓ clean (“situational utility rises”; Brock–Durlauf)                                                                      | defensible, but against the grain of the very citation it borrows (§5.5)                                                                  |
| Quality $\times$ social proof | substitutes (social contribution independent of $\Gamma$ )                                                                | <b>complements</b> ( $\partial^2 U / \partial \Gamma \partial \tilde{A} > 0$ ) — strong assumption, but interesting and testable          |
| Immunity                      | $q_i > \Gamma + \lambda$ , degree-independent (§5.1)                                                                      | $q_i > \Gamma k_i^\gamma$ , <b>degree-dependent</b> (averse + poorly connected) — arguably more realistic                                 |
| Empirical status              | ✓ <b>already validated</b> : replicates the current model’s heatmaps; $\lambda \approx 0.7$ identified unimodally         | not yet run; $\gamma$ to be calibrated with the same procedure                                                                            |
| Hub behavior                  | bounded ( $\tilde{E} \leq 1$ )                                                                                            | at high $\gamma$ , a saturated hub receives reinforcement $\sim k^\gamma$ — may over-fire on dense networks                               |

**Recommendation.** Use the **linear-fractional variant as the primary operationalization** (validated, Valente-compatible, microeconomically clean,  $\lambda$  identified from the legacy data), and the  **$\gamma$ -form as the formal bridge to Tur and as a robustness check**: present the exact nesting (§4.6) in the text, and run a short  $\gamma$  sweep — calibrated with our own fit procedure, never by borrowing Tur’s numbers — to show that the conclusions do not hinge on the additive-vs-multiplicative composition. This keeps the best of both: the tradition and the validation of the linear form, and the exact nesting as an argument, without inheriting  $\gamma$ ’s transportability trap.

## 6 Side-by-side with the current model and Tur

|                                        | NDFU current                                                     | Tur 2024                   | Unified rule (both variants)                                                                            |
|----------------------------------------|------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------|
| Decision rule                          | two channels (OR)                                                | single utility rule        | <b>single utility rule</b>                                                                              |
| Threshold source                       | exogenous<br>$\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ | emerges from reinforcement | <b>derived from measured <math>q_i</math>:</b><br>$(q_i - \Gamma)/\lambda$ or $(q_i/\Gamma)^{1/\gamma}$ |
| Preferences modified during diffusion? | no                                                               | yes ( $u$ lowered)         | <b>no</b> under the situational-utility framing (§4.2, §5.5)                                            |
| Distance enters via                    | stochastic gate in $\tilde{E}$                                   | — (absent)                 | <b>same stochastic gate</b> , in $\tilde{E}$ or in $\tilde{A}$                                          |
| Heterogeneity source                   | $q$ empirical + $\tau$ artificial                                | $u$ artificial             | <b><math>q</math> empirical only</b>                                                                    |
| Free params (beyond $\Gamma, h$ )      | $\tau_{\text{mean}}, \tau_{\text{sd}}$                           | $\gamma$                   | $\lambda$ (linear) or $\gamma$ ( $\gamma$ -form)                                                        |
| Nests Tur 2024?                        | no                                                               | —                          | <b>exactly via the <math>\gamma</math>-form</b> (§4.6); linear form only qualitatively                  |
| netdiffuseR as-is?                     | yes                                                              | no                         | <b>yes</b> (fractional or count threshold.dist)                                                         |
| Main risk                              | Bianchi critique                                                 | distance-blind             | heatmaps shift with the $\tau$ - $q$ coupling (checked: they don't, §7)                                 |

## 7 Calibration notes and empirical status

- Comparability anchor: the main-text configuration of the current engine is  $\tau \sim \mathcal{N}(0.40, 0.12)$ . Under the linear-fractional variant,  $\tau_i^* = \max\{0, (q_i - \Gamma)/\lambda\}$ , so the implied threshold distribution *depends on  $\Gamma$  and the empirical  $q$  distribution*. The sweep script reports, per  $(\lambda, \Gamma)$ : mean and SD of  $\tau^*$ , share of instigators ( $\tau^* = 0$ ), and share of immunes ( $\tau^* > 1$ ).
- **Empirical status (linear variant).** An initial minimal test (12 networks, coarse grid; script since removed) showed the qualitative phase structure survives. The extended sweep (10\_unified\_model\_sweep.R: 96 GSS networks, the engine's full  $41 \times 13$  grid,  $\lambda \in \{0.3, 0.4, \dots, 1.5\}$ , random seeding, 665,184 simulations) found a clean **unimodal best fit to the current model at  $\lambda \approx 0.7$**  ( $r = 0.94$  total, 0.98 rational, 0.93 social against the  $\tau \sim \mathcal{N}(0.40, 0.12)$  surface;  $\lambda \in [0.6, 0.9]$  all  $r > 0.91$ ). Known differences: tipping one MSP-grid step earlier, and the immunity ceiling at low  $\lambda$  (§5.1).
- On the  $\lambda$  range:  $\{0.3, \dots, 1.5\}$  was a pragmatic bracketing (the minimal test showed 0.5 and 1.0 straddling the current model and 2.0 clearly too strong); the sweep's unimodal peak with degradation at *both* flanks justifies the endpoints ex post. Theoretical landmark:  $\lambda = 1$  is the **immunity boundary** — since  $q \leq 1$ ,  $\lambda \geq 1$  implies zero structurally-immune agents at every  $\Gamma > 0$ , while  $\lambda < 1$  leaves an immune share  $\Pr(q > \Gamma + \lambda)$ . There is no theoretical bound at  $\lambda = 1$  ( $\lambda$  is a utility contribution, like Brock–Durlauf's unbounded  $J$ ); production runs use the calibrated band  $[0.5, 0.9]$ .
- Still pending: the mirror  $\gamma$ -form sweep (count threshold  $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$  on the same machinery,  $\gamma$  calibrated by the same fit procedure), the remaining seeding strategies, and the ATP/innovation system.

### 7.1 The structural premium survives the unified rule

Run by scripts/05\_unified\_diffusion\_sweep.R: GSS vs. genuine degree-preserving twins (igraph::rewire(k built fresh and persisted by scripts/04\_make\_DP\_networks.R),  $\lambda \in \{0.5, \dots, 1.0\}$ , 41 IUL levels

× a denser **25-level MSP grid** (step 1/24, nesting the legacy 13-level grid), 96 networks, random seeding: 1,180,800 simulations. Analysis in `scripts/07_premium_sensitivity.R`.

| $\lambda$                                               | 0.5    | 0.6            | 0.7    | 0.8    | 0.9    | 1.0    |
|---------------------------------------------------------|--------|----------------|--------|--------|--------|--------|
| $\beta_{\text{DP}}$ (odds)                              | −0.642 | − <b>0.859</b> | −0.960 | −1.227 | −1.251 | −1.261 |
| odds drop                                               | −47%   | − <b>58%</b>   | −62%   | −71%   | −71%   | −72%   |
| $\bar{\Phi}_{\text{GSS}} - \bar{\Phi}_{\text{DP}}$ (pp) | 4.8    | 5.5            | 5.7    | 7.7    | 6.9    | 7.9    |
| % of plane with gap > 0.10                              | 18.9   | 21.1           | 21.9   | 25.6   | 23.4   | 23.8   |
| mean gap inside that band (pp)                          | 17.8   | 19.8           | 20.6   | 23.0   | 24.6   | 26.5   |
| $MSP_c$ shift (DP − GSS)                                | 0.061  | 0.068          | 0.069  | 0.072  | 0.065  | 0.071  |

**Reading.** The premium survives, and at the calibrated coupling it reproduces the legacy engine almost exactly:  $\lambda = 0.6$  gives  $\beta_{\text{DP}} = -0.859$  against the legacy  $-0.850$ .

But the  $\lambda$ -dependence must be read with care, because the two scales disagree:

- On the **odds** scale the premium nearly doubles ( $-0.64 \rightarrow -1.26$ ) and then *plateaus* from  $\lambda \approx 0.8$  (steps per decile:  $-0.217, -0.100, -0.267, -0.024, -0.011$ ).
- On the **probability** scale it is roughly flat and non-monotonic (4.8–7.9 pp).

The gap between the two is a link artifact: as  $\lambda$  grows, both topologies push more of the plane into saturation (the share of cells with  $\Phi > 0.95$  rises from 51% to 72% for GSS), and in the saturated regime the logit converts a fixed probability gap into an ever larger log-odds gap. **The growth of  $\beta_{\text{DP}}$  with  $\lambda$  therefore measures saturation, not extra structural separation.**

**The  $\lambda$ -robust statements** (recommended for the paper):

1. **Where it lives.** The premium is concentrated in an L-shaped band along the tipping frontier: 19–26% of the (IUL, MSP) plane, where GSS gains 18–27 percentage points. Outside that band the two topologies agree (both dead, or both saturated). Structure decides where the system is undecided.
2. **Tipping shift.** DP needs about *twice* the social openness to tip ( $MSP_c \approx 0.065$ – $0.077$  for GSS vs.  $0.13$ – $0.145$  for DP), and the shift varies by only  $\pm 8\%$  across the whole  $\lambda$  band — the most stable way to state the premium. It echoes the legacy contrast ( $MSP_c = 0.25$  vs.  $0.417$ ).
3. **Channel split.** The rational channel is flat in  $\lambda$  ( $\beta \approx -0.22$ ) while the social channel carries the growth ( $-0.21 \rightarrow -0.37$ ). Note that here the rational channel is *not* topology-blind: its adopters still need one gate-passing contact, and DP ties are socially more distant (mean Gower  $0.229 \rightarrow 0.325$ ), so the gate rejects more often. In the legacy engine the analogous coefficient was a second-order accounting artifact; here it is a mechanism.

**A reverse-premium corner (worth reporting honestly).** At  $\lambda = 0.5$ , with high openness ( $h \geq 0.75$ ) and moderate attractiveness ( $\Gamma \in [0.20, 0.275]$ ), **DP beats GSS** by up to 5.7 pp (3.4% of the plane); the effect vanishes for  $\lambda \geq 0.7$ . The mechanism is Tur (2024)’s own: with the gate wide open, homophily *isolates the resistant among the resistant*, whereas random mixing seats early adopters next to reluctant agents and converts them. That the new engine reproduces a published theoretical mechanism it was not built to produce is a validity check — and it makes the premium claim properly conditional rather than universal.

## 8 To verify before writing any of this into the paper

- ⊠ Exact reinforcement functional form in Tur 2024:  $u_t = u_0 \cdot (1/a_t)^\gamma$  (multiplicative, power-law). *Resolved against both PDFs (2026-08-31):* Tur 2018 (PRE), Eq. (1), uses the **same**

form (there called MQR,  $q_t^i = q_0^i(1/a_t^i)^\gamma$ );  $a_t^i \in \mathbb{N}$  is explicitly the number of **adopting neighbors** (a state count, not received recommendations), with the rule evaluated only once informed ( $a \geq 1$ ). Both papers restrict  $\gamma \in [0, 1]$  — justified in 2018 by non-increasing marginal social influence (concavity), with the reductio that large  $\gamma$  trivializes diffusion on their  $k = 4$  lattices — and sweep  $\gamma$  as a free “intensity” parameter (2024, Fig. 5:  $\gamma \in \{0, 0.5, 1\}$ ; heatmaps over  $[0, 1]$ ). **No empirical calibration of  $\gamma$  exists in either paper**; see the transportability caveat in §5.3.

- Whether Tur 2024’s homophily-rewiring  $\rho$  has a formal analogue we should cite when claiming nesting (their  $\rho \approx$  our imputation; not identical constructs).
- The random-utility framing sentence for IUL (already drafted in JOURNAL\_FIT\_ANALYSIS.md, §4d) composes cleanly with the situational-utility framing:  $U_i(t) = \Gamma + \lambda \tilde{E}_i$  is a private valuation plus a social term — this is Brock & Durlauf (2001)’s decomposition (§5.6), worth citing.